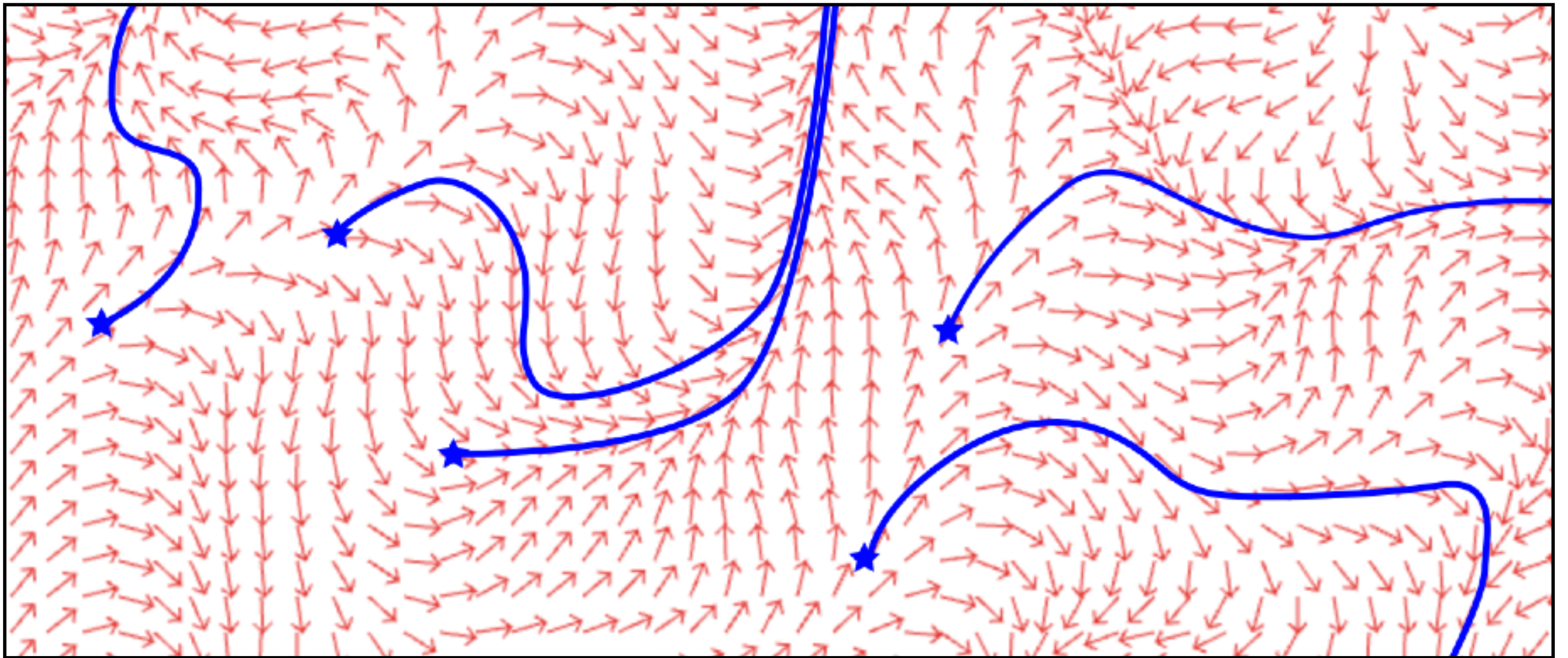


Mean-field Dynamics



$$\partial_t X_t = v(t, X_t, \mu_t) \quad (\text{LAG})$$

$$\partial_t \mu_t = \operatorname{div}_{\mathcal{M}}(v(t, \cdot, \mu_t) \mu_t) \quad (\text{EUL})$$

Geometry of Particle Methods

Jordan, Kinderlehrer, Otto 1998: reinterpret Langevin dynamics as a *minimizing movement* scheme:

$$\mu_{k+1} := \arg \inf_{\mu} \text{KL}(\mu, \pi) + \frac{1}{2\eta} W_2^2(\mu, \mu_k) \quad \pi, \mu_k \in \mathcal{P}(\mathbb{R}^d)$$

First insights into rich theory of *Wasserstein Geometry*, a “Riemannian” calculus on $\mathcal{P}(\mathbb{R}^d)$!